

Credit - Drishti Ias

EOS-05: India's 1st Dedicated Geosynchronous Earth Observation Satellite



Source: TH

Why in News?

The **Indian Space Research Organisation (ISRO)** successfully launched the **Earth Observation Satellite-05 (EOS-05)** aboard the **GSLV-F17** launch vehicle from the Satish Dhawan Space Centre, Sriharikota. This marks a major milestone as it is India's **first dedicated imaging satellite** designed to operate from a **geosynchronous orbit**.

What is the EOS-05 Mission?

- **About the Satellite:** Dubbed the “**Eye in the Sky**,” EOS-05 is a **state-of-the-art Earth observation spacecraft** equipped with advanced optical and infrared imaging payloads to provide **continuous, real-time imagery**.
 - Weighing **2,367 kg**, it represents the **heaviest payload** successfully injected into orbit by a standard GSLV configuration.
- **The Launch Vehicle:** The mission was executed by **GSLV-F17**, marking the **19th operational flight** of India's Geosynchronous Satellite Launch Vehicle.
 - The **GSLV is a three-stage launch vehicle**. The **first stage** uses a solid core with **four liquid strap-on boosters**, followed by a **liquid second stage** and an indigenous **cryogenic upper stage (CUS)** for launching heavier payloads into deep transfer trajectories.
- **The Orbit:** Unlike conventional **Earth observation satellites** that circle close to the planet in Low Earth Orbit (LEO) with limited revisit windows, EOS-05 is **India's first dedicated imaging satellite designed to operate from a Geosynchronous Orbit (GEO)**, roughly **36,000 km above the Earth**.
- **Mission Profile & Orbit Raising:** GSLV-F17 precisely injected the spacecraft into a **Sub-Geosynchronous Transfer Orbit (Sub-GTO)**.
 - The satellite will progressively fire its onboard liquid apogee motor (LAM) over multiple orbit-raising maneuvers to reach its **designated geostationary orbital slot over the Indian subcontinent**.
- **Significance of Imaging Satellites in GEO:** LEO imaging platforms pass over a specific area in minutes, leaving blind spots of hours or days before the next satellite passes. GEO provides **continuous persistence** and an unblinking, wide-area view.
 - GEO imaging enables persistent, high-frequency tracking of rapidly evolving meteorological hazards,

such as **cloudbursts, cyclones, flash floods** , and **forest fires** , where hourly updates are vital for disaster relief.

Key Applications of EOS-05

Continuous observation.
A safer, more secure and sustainable India.

From Earth Insights to a **Better Tomorrow**

- Continuous Monitoring
- Real-time Insights
- Stronger Security
- Sustainable Future

Disaster Management Faster Response, Safer Lives	Strategic & National Security Stronger Borders, Safer Nation	Agriculture & Environment Healthy Ecosystems, Prosperous Tomorrow
Enables near real-time monitoring of cloudbursts, flash floods and cyclones, supporting rapid emergency response.	Provides continuous surveillance of India's land borders and maritime domain, enhancing situational awareness.	Supports monitoring of forest fires, vegetation health and crop cycles through continuous observation.
 Cyclone Tracking	 Land Border Surveillance	 Forest Fire Monitoring
 Flood Monitoring	 Maritime Domain Awareness	 Vegetation Health Assessment
 Cloudburst Alerts		 Crop Cycle Monitoring
Early warnings. Quicker action. Reduced human and economic losses.	Greater vigilance. Enhanced situational awareness. A more secure India.	Healthier ecosystems. Smarter farming. A more sustainable future.

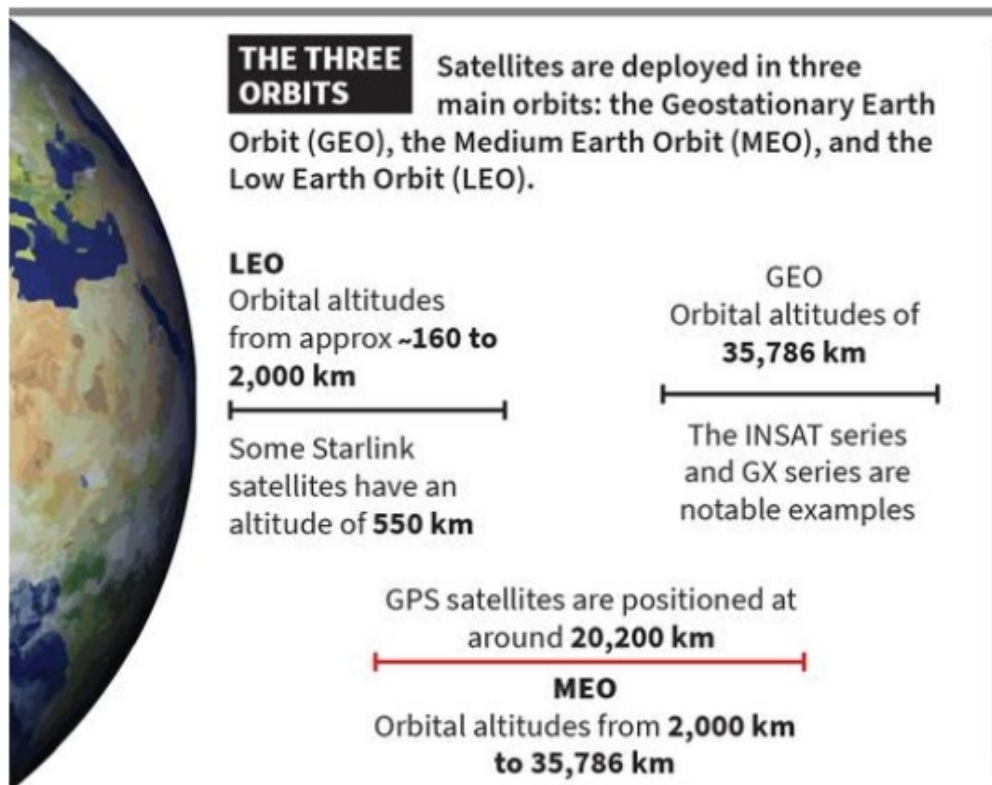
People | Prosperity | A Safer Planet

EOS-05: Observing Today for a Brighter Tomorrow

What are the Different Types of Satellite Orbits?

- **Orbit:** According to the **European Space Agency (ESA)** , an **orbit** is the curved path an object follows around another body under the influence of gravity. Earth orbits are broadly classified into three altitude classes:
- **Low Earth Orbit (LEO):**
 - **Altitude & Speed:** Ranges from **160 km to 2,000 km** above Earth's surface. Satellites complete an orbit in **90 to 120 minutes** , traveling at high velocities to counteract gravitational pull.
 - **Applications:** Proximity minimizes **signal latency** , making LEO ideal for high-resolution imaging, broadband mega-constellations, and crewed platforms like the **International Space Station (ISS)** .
 - **Subsets:** Includes **Sun-Synchronous Orbits (SSO)** (consistent solar illumination) and **Very Low Earth Orbit (VLEO)** (ultra-high resolution, but prone to high atmospheric drag and fuel expenditure).
 - **Density:** As of late 2026, over **15,500 active satellites** operate in LEO, with India maintaining **22 operational satellites** in this band.
- **Medium Earth Orbit (MEO):**
 - **Altitude:** Extends from **2,000 km up to 35,786 km** , positioned directly beneath the geosynchronous belt.
 - **Characteristics:** Strikes a balance between low signal latency and expansive ground coverage.
 - **Applications:** Primarily hosts **Global Navigation Satellite Systems (GNSS)** such as the European Union's **Galileo constellation** (30 satellites) and the US **GPS network**.
- **Geosynchronous Orbit (GSO) & Geostationary Orbit (GEO):**
 - **Altitude & Motion:** Positioned at approximately **35,786 km** . Satellites orbit eastward matching Earth's rotational period of **23 hours, 56 minutes, and 4 seconds** .
 - **The GEO Distinction:** When an orbit has a **zero-degree inclination directly over the equator** , it is a **Geostationary Orbit (GEO)** .
 - To an observer on the ground, the satellite appears permanently fixed at a single point in the sky.

- **Footprint:** A single GEO satellite observes nearly one-third of the globe; **three evenly spaced satellites can deliver near-global coverage** .



Frequently Asked Questions (FAQs)

1. What is the significance of EOS-05?

EOS-05 is designed for **continuous Earth observation from geosynchronous orbit** , supporting disaster management, surveillance, agriculture and environmental monitoring.

2. What is the difference between Geosynchronous and Geostationary Orbits?

A **geosynchronous satellite** has an orbital period equal to Earth's rotation, while a **geostationary satellite** is a special geosynchronous satellite that remains apparently fixed over the equator.

3. What are the major applications of EOS-05?

Its applications include **disaster monitoring, border and maritime surveillance, forest-fire detection, vegetation assessment and crop-cycle monitoring** .

4. What are the major features of GSLV?

GSLV is a three-stage launch vehicle using a solid first stage, liquid second stage and an indigenous **cryogenic upper stage** , enabling it to launch heavier payloads.

5. Why is continuous Earth observation important?

It enables **near real-time monitoring of rapidly changing events** , improving early warning, emergency response, national security and environmental management.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/po8kIKZvk1w](https://www.youtube.com/embed/po8kIKZvk1w)]

UPSC Civil Services Examination, Previous Year Question (PYQ)

Prelims

Q. With reference to India's satellite launch vehicles, consider the following statements: (2018)

1. PSLVs launch the satellites useful for Earth resources monitoring whereas GSLVs are designed mainly to launch communication satellites.
2. Satellites launched by PSLV appear to remain permanently fixed in the same position in the sky, as viewed from a particular location on Earth.
3. GSLV Mk III is a four-staged launch vehicle with the first and third stages using solid rocket motors; and the second and fourth stages using liquid rocket engines.

Which of the statements given above is/are correct?

- (a) 1 only
- (b) 2 and 3
- (c) 1 and 2
- (d) 3 only

Ans: (a)

Mains

Q. What is the main task of India's third moon mission which could not be achieved in its earlier mission? List the countries that have achieved this task. Introduce the subsystems in the spacecraft launched and explain the role of the 'Virtual Launch Control Centre' at the Vikram Sarabhai Space Centre which contributed to the successful launch from Sriharikota. **(2023)**

Q. What is India's plan to have its own space station and how will it benefit our space programme? **(2019)**

Q. Discuss India's achievements in the field of Space Science and Technology. How the application of this technology helped India in its socio-economic development? **(2016)**

Article 142 of the Constitution



For Prelims: Article 142 of the Constitution , First Information Report , Supreme Court of India , Judicial activism and judicial restraint .

For Mains: Article 142 and Complete Justice, Judicial Activism vs Judicial Restraint, Separation of Powers

Why in News?

The **Supreme Court of India** invoked its extraordinary powers under **Article 142 of the Constitution** to quash multiple **First Information Reports (FIRs)** filed across various states against participants of the nationwide examination protests led by the **Cockroach Janta Party (CJP)**.

- The Court's order gave legal force to a negotiated settlement between the **Union government and the protest group over the NEET-UG 2026 exam leak**.
- By clubbing and scrapping these cases directly, the Supreme Court bypassed the conventional statutory route for **quashing FIRs, sparking a fresh debate on the scope, application, and boundaries of Article 142**.

Summary

- Article 142 empowers the Supreme Court to ensure "complete justice", but its broad and undefined scope raises concerns about judicial overreach, excessive discretion and separation of powers.
- The way forward lies in judicial restraint, adherence to precedent, reasoned orders and stronger institutional safeguards to ensure Article 142 supplements rather than bypasses statutory law.

What is Article 142 of the Constitution?

- **Defining the Power:** Article 142(1) empowers the Supreme Court to pass any "**decree or order as it is necessary for doing complete justice in any cause or matter pending before it.**"
 - The Constitution notably does not define what constitutes "complete justice" or the specific circumstances for its use.
 - Article 142(2) empowers the Court to secure the attendance of persons, discovery or production of documents, investigation, and punishment for contempt, subject to laws made by Parliament.
- **Historical Origins:** The provision traces its roots to **Section 210 of the Government of India Act, 1935**, and earlier colonial-era practices of resolving disputes based on "**justice, equity and good conscience**" when written law was absent.
- **Constituent Assembly Adoption:** Originally drafted as Article 118, it was adopted by the Constituent Assembly in 1949, without any formal debate. Two proposed amendments were moved but subsequently withdrawn.
- **Undefined Scope:** Because it was passed without debate, the Constituent Assembly left the definition of this expansive power open, allowing the Supreme Court to interpret and shape its boundaries on a case-by-case basis over the past seven decades.
- **Significant Historical Invocations of Article 142:**
 - **Frequent Invocation:** Because of its open-ended nature, Article 142 has become one of the Court's most utilized tools. An IIM Ahmedabad study found 1,579 references and 791 direct invocations of this power between 1950 and 2023, primarily in civil matters.

- **Bhopal Gas Tragedy (1989):** In the [Union Carbide Corporation v. Union of India case](#), the Supreme Court invoked Article 142 to approve a USD 470 million settlement between the parties and quash the criminal proceedings against the corporation.
- **Vishaka Guidelines (1997):** In the absence of parliamentary legislation on **workplace sexual harassment**, the Supreme Court invoked **Article 142** to lay down the [Vishaka Guidelines in Vishaka v. State of Rajasthan \(1997\)](#), which remained binding until Parliament enacted the [Sexual Harassment of Women at Workplace \(Prevention, Prohibition and Redressal\) \(PoSH\) Act, 2013](#).
- **Ayodhya Title Dispute (2019):** The power was used to settle the long-standing dispute on a "preponderance of probability" and to direct that five acres of land be allotted to the Sunni Waqf Board.
- **A.G. Perarivalan vs The State Of Tamil Nadu State (2022):** The Court invoked Article 142 to order the release of convict A.G. Perarivalan.
- **Matrimonial Disputes:** The Court regularly uses this power to grant divorces in cases of irretrievable breakdown of marriage, bypassing standard statutory waiting periods, and routinely applies it in public interest litigations.
- **SC Attempts to Limit Powers under Article 142:** The Court has made several internal attempts to establish judicial guardrails to prevent undue discretion.
 - **Consistent with Fundamental Rights:** In *Prem Chand Garg vs. State of Uttar Pradesh (1962)*, the Court ruled that orders made under Article 142 must align with fundamental rights and existing substantive law.
 - **Supplementing, Not Supplanting:** In *Supreme Court Bar Association vs. Union of India (1998)*, a Constitution Bench clarified that Article 142 is intended to **supplement a statute by filling gaps, not to supplant or override an existing statutory framework**.
 - Despite these self-imposed limitations, the Court has frequently ignored its own cautionary precedents.

What are the Concerns Regarding the SC's Exercise of Powers under Article 142?

- **Circumventing Statutory Schemes:** In the CJP protest order, thousands of FIRs were clubbed and quashed without individual scrutiny.
 - Ordinarily, quashing an FIR must be examined on its own facts under **Section 528 of the Bharatiya Nagarik Suraksha Sanhita (BNSS), 2023** to see if an offence is made out. Critics argue that utilizing Article 142 for mere "**convenience**" **supplants the legislative procedure**.
- **Undefined and Unfettered Discretion:** The absence of a precise definition of "complete justice" under Article 142 gives the Supreme Court wide discretionary powers. The provision has therefore been criticised for potentially enabling broad and unchecked judicial discretion.
- **Magnification in a Polyvocal Court:** The Supreme Court consists of 38 judges who **largely sit in Division Benches of two**. When undefined discretion is wielded by smaller benches, **judicial outcomes can heavily depend on the individual philosophy** of a specific two-judge bench rather than institutional consistency.
- **Judicial Overreach:** The Supreme Court has sometimes been criticised for entering areas traditionally falling within the **legislative or executive domain**, raising concerns about [separation of powers](#), a part of the [basic structure](#) of the Constitution.
 - **Highway Liquor Ban:** The Court invoked **Article 142** to prohibit liquor shops within **500 metres of**

National Highways and extended the restriction to **State Highways** , despite the absence of similar notifications by state governments.

- Such interventions raise concerns about the **extent of judicial discretion under Article 142** and the balance between **judicial activism and judicial restraint** .
- **Lack of External Checks:** The limitations on Article 142 are entirely self-imposed by the judiciary. If a bench disregards the limits set by previous Constitution benches, no institution outside the Court can enforce those guardrails.

Global Perspective: Extraordinary Judicial Powers

- **South Africa: Section 172(1)(b) of the Constitution** empowers courts to make “**any order that is just and equitable**” when deciding constitutional matters, providing a constitutional basis for remedial judicial intervention.
- **Canada:** Courts can grant “**appropriate and just**” remedies under **Section 24(1) of the Canadian Charter of Rights and Freedoms** for violations of constitutional rights.

Way Forward

- **Larger Benches for Invocation:** As suggested by legal luminaries, the power to invoke Article 142 could be restricted to Constitution Benches of at least five judges to ensure collective wisdom and consistency.
- **Mandatory Reasoning on Record:** The Court should ensure that every exercise of **Article 142** is supported by **reasoned orders and the principle of proportionality** , clearly explaining the statutory gap being addressed and why the intervention is necessary.
- **Strict Adherence to Precedent:** The judiciary must institutionalize the principles laid down in the *Supreme Court Bar Association* case, ensuring that Article 142 is strictly used as a curative tool of last resort, and not as a shortcut to bypass established criminal or civil procedure.

Drishti Mains Question:

Article 142 enables the Supreme Court to do “complete justice”, but its undefined scope raises concerns about judicial overreach. Critically examine.

Frequently Asked Questions (FAQs)

1. What does Article 142 of the Constitution provide?

It empowers the **Supreme Court to pass orders necessary to do “complete justice”** in any matter pending before it.

2. Can Article 142 override existing statutory law?

In *Supreme Court Bar Association v. Union of India (1998)*, the Court held that Article 142 **supplements rather than supplants statutory law**.

3. What limitation was established in *Prem Chand Garg v. State of Uttar Pradesh (1962)*?

The Supreme Court held that orders under Article 142 must remain **consistent with Fundamental Rights and substantive law**.

4. How are FIRs ordinarily quashed under the BNSS?

Section 528 of the Bharatiya Nagarik Suraksha Sanhita (BNSS) provides the High Courts with inherent powers to prevent abuse of process and secure the **ends of justice**.

5. What are the major concerns surrounding Article 142?

Key concerns include **judicial overreach, excessive discretion, circumvention of statutory procedures, separation-of-powers concerns and inconsistent application**.

[Watch Video on YouTube: [▶ https://www.youtube.com/embed/8tliPt3L8Hc](https://www.youtube.com/embed/8tliPt3L8Hc)]

UPSC Civil Services Exam, Previous Year Questions (PYQ)

Q. With reference to the Constitution of India, prohibitions or limitations or provisions contained in ordinary laws cannot act as prohibitions or limitations on the constitutional powers under Article 142. It could mean which one of the following? (2019)

(a) The decisions taken by the Election Commission of India while discharging its duties cannot be challenged in any court of law.

(b) The Supreme Court of India is not constrained in the exercise of its powers by laws made by the Parliament.

(c) In the event of a grave financial crisis in the country, the President of India can declare a Financial Emergency without the counsel from the Cabinet.

(d) State Legislatures cannot make laws on certain matters without the concurrence of the Union Legislature.

Ans: (b)

Mains

Q. Do you think that the Constitution of India does not accept the principle of strict separation of powers rather it is based on the principle of ‘checks and balance’? Explain. (2019)

Mount Anak Krakatau



Source: TOI

Indonesia's **Mount Anak Krakatau erupted** , releasing **fine volcanic ash particles** that disrupted air transport by creating risks to aircraft engines, reducing visibility, and posing health hazards, leading to suspension of operations at several airports and affecting passengers and flights.

- **About:** Mount Anak Krakatau is located in the **Sunda Strait between Java and Sumatra islands of Indonesia** . It is among Indonesia's **around 130 active volcanoes** and lies within the **Pacific Ring of Fire** , a highly tectonically active region associated with frequent earthquakes and volcanic eruptions.
 - The volcano is **part of the Sunda Arc**, formed due to the **subduction of the Indo-Australian Plate beneath the Eurasian Plate** , making the region highly vulnerable to volcanic activity.
- **Origin:** Anak Krakatau, meaning **"Child of Krakatau,"** emerged from the remains of the **Krakatau volcanic system** after the catastrophic **1883 eruption** , which collapsed much of the original island, triggered devastating tsunamis, and caused over **36,000 deaths** .
- **Significance:** Anak Krakatau highlights the importance of **volcanic monitoring, disaster preparedness, and tsunami early warning systems** , as demonstrated by the 2018 flank collapse that triggered a tsunami in the Sunda Strait.



Read More: [Volcanism](#)

Bhutan Achieves Rabies Elimination Milestone



Source: TH

Bhutan became the 1st country in the WHO South-East Asia Region to eliminate **dog-transmitted rabies as a public health problem**, maintaining zero human deaths since June 2023 through surveillance, vaccination, and a **One Health approach**.

- The validation was conducted as per the **World Health Organization (WHO) Expert Consultation on Rabies (2018) criteria**, with assessment by experts supported by the **WHO and the World Organisation for Animal Health (WOAH)**, covering surveillance systems, laboratories, records, and field implementation.
- Elimination of **dog-mediated human rabies as a public health problem** requires **no human deaths from dog-transmitted rabies for at least two years** along with effective **surveillance, vaccination, and control mechanisms** to prevent re-emergence.

Rabies

- **About:** Rabies is caused by the **rabies virus, a neurotropic virus** belonging to the **Lyssavirus genus**, which affects the central nervous system and leads to fatal encephalitis if untreated after symptom onset.
- **Nature of the disease** : Rabies is a **viral, zoonotic disease and a neglected tropical disease (NTD)**. Once clinical symptoms appear, rabies is almost invariably fatal.
- **Transmission:** Rabies spreads through the **saliva of infected animals**, mainly through **bites and scratches**, or contact with broken skin or mucous membranes. **Human-to-human transmission** is extremely rare and has mainly been reported through organ transplantation.
- **Prevention and Treatment** : Rabies deaths are **completely preventable** with timely **post-exposure prophylaxis (PEP)**. PEP includes:
 - Immediate wound washing with soap and water (15 minutes)
 - A full course of **rabies vaccine**
 - **Rabies immunoglobulin (RIG)** or monoclonal antibodies for severe exposures
- **Symptoms:** Incubation period usually **2-3 months** (can range from one week to one year). **Once clinical symptoms appear, rabies is virtually 100% fatal.**
 - Two major forms include **Furious rabies** (hydrophobia, hallucinations, hyperactivity, rapid death) & **Paralytic rabies** (gradual paralysis, often misdiagnosed).
- **Rabies Burden:** India continues to face a significant rabies burden, largely linked to dog bites, while the disease imposes an **estimated global economic cost of US\$8.6 billion annually**, including healthcare expenses, livelihood losses, and the financial burden of **post-exposure prophylaxis (PEP)**.

Read More: [Rabies in India](#)

